

# MARCO ZANOTTI

I am a data scientist with broad knowledge in time series analysis, forecasting, anomaly detection, and econometrics.

I specialized in developing and enhancing business forecasting processes and practices. Over the years I developed and contributed to core forecasting systems in sectors like e-commerce fashion, airline, tourism and travel. I also guided and mentored junior colleagues in developing predictive models that enable accurate forecasting.

I am passionate about advancing the field of forecasting and fostering collaboration between academia and industry. I do academic work in forecasting, and I am a member of the International Institute of Forecasters. Recently I have enjoyed teaching professionals and university students about time series analysis, machine learning, programming, and statistics.



## PROFESSIONAL EXPERIENCE

- current  
|  
2026

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**Senior Data Science Engineer, Forecasting Specialist**  
Sabre - Airline - Full remote  Dallas, USA

- Developed demand forecasting solutions for airline clients in revenue management, applying forecasting methodologies to support pricing optimization decisions.
- 2026  
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2026

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**Senior Researcher, Forecasting Specialist**  
iqast - Software - Full remote  Hamburg, Germany

- Led demand forecasting analysis on global container shipping, focusing on model retraining strategies to maintain forecast accuracy and stability through structural market breaks.
  - Contributed, with the research team, to improve the company's forecasting software
- 2025  
|  
2024

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**Senior Data Scientist, Forecasting Specialist**  
Wanan Luxury - E-commerce Fashion - Full remote  Rome, Italy

- Led the company's time series forecasting projects
  - Coordinated and contributed to the development of a forecasting tool to predict future demand for tens of thousands item-level (SKUs) time series across several markets worldwide, and with different product hierarchies, resulting in increased forecasting accuracy and efficiency
  - Integrated probabilistic methods to support more informed decisions
  - Designed and developed an app for evaluating forecasting experiments, enhancing reliability of the analyses
  - Provided mentorship to junior colleagues, supporting their professional growth and development in the field of forecasting
- 2024  
|  
2021

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**Data Scientist**  
Blogmeter - Digital Services - Hybrid  Milan, Italy

- Integrated machine learning solutions into company's applications
  - Developed an anomaly detection solution to promptly communicate relevant events to the clients
  - Coordinated the Product team

## Contacts

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-  [gh-marcozanotti](#)
-  [in lk-marcozanotti](#)
-  +39 334 9808108

## Core Skills

High experience in time series modeling and forecasting, ML modeling and text mining

## Languages



2021  
|  
2020



### Data Scientist

T-Voice - Digital Services - Full remote

📍 Milan, Italy

- Contributed to the development of an AI platform for the analysis of textual data
- Effectively incorporated text data into forecasting models
- Coordinated the R&D team

2020  
|  
2017



### Junior Data Scientist

Uvet Amex GBT - Tourism and Aviation - Hybrid

📍 Milan, Italy

- Developed and maintained diverse statistical and machine learning models to predict: sales at different aggregations to support strategic decisions; daily inbound calls to optimize scheduling hours; air passenger demand over several destinations to support aircraft planning; business car rental demand to support fleets management
- Successfully improved the approach to forecasting, resulting in a significant reduction of errors and facilitating the faster adoption of machine learning solutions by the business
- Implemented an anomaly detection system to signal possible congestions in the company's reservation platform
- Developed econometric models to measure key drivers of business travel and tourism
- Supported automation processes implementing OCR solutions



## TEACHING EXPERIENCE

current  
|  
2019



### Adjunct Professor

University of Milan - Hybrid

I teach forecasting, machine learning, coding, and statistics.

current  
|  
2019



### Business Consultant

Business Consultancy - Hybrid

I offer consultancy and training about forecasting, machine learning, and the development of business specific statistical models.



## AWARDS

2025



### IIF-SAS Forecasting Research Grant

International Institute of Forecasters - SAS

2025



### International Symposium on Forecasting Award

International Institute of Forecasters

📍 Beijing, China

## Programming

R

Python

Bash

## BI & Visualization

Shiny

## Databases

SQL

InfluxDB

MySQL

PostgreSQL

Big Query

## Versioning

Git

GitHub

GitLab

BitBucket

## CI / CD

GitHub Actions

Codecov

Jenkins

## Cloud

Google Cloud Platform

Amazon Web Services

## Project Management

Jira

Asana

## Operating Systems

Linux



## EDUCATION

2026  
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2022

- **Ph.D., Statistics**  
University of Milano-Bicocca  Milan, Italy

Research on time series forecasting, cross-learning, forecasting stability, time series foundation models (LLMs), MLOps, and green AI

2018

- **Post-graduate, Data Science for Economics, Business and Finance**  
University of Milan  Milan, Italy

2017  
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2015

- **M.Sc., Economics and Finance**  
University of Milan  Milan, Italy

2014  
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2011

- **B.Sc., Political Sciences and International Relations**  
University of Parma  Parma, Italy  
ERASMUS, University of Lapland, Rovaniemi, Finland



## PERSONAL COMPETENCES

- **Organisational and Management**

- Project management: I coordinated the development of different projects, from analysis to implementation
- People management: I coordinated the activities of junior colleagues, supervising and guiding in their professional development
- High relational and communication skills
- High resistance to stress
- Since 2020 I proficiently work from remote

## Certificates

- [Certified Forecaster and Demand Planner](#)
- [High-Performance Time Series Forecasting](#)
- [Designing Forecasting Pipelines for Production](#)
- [Time Series Analysis](#)
- [Deep Learning for Sequence Modelling and Times Series](#)
- [Python for Machine Learning and APIs](#)
- [Python for Data Science Automation](#)
- [Shiny Web Applications](#)
- [Macroeconometric Forecasting](#)

## Sports & Hobbies

I was a footballer. I played in Italian minor championships until 22. Now I practice kitesurf, an amazing waterboard sport, and I do canicross with my little friend Milo.

I like manga, boardgames, videogames, curling and footgolf. I love listening folk and country music.

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*Last updated on July 2026*



## PUBLICATIONS

- 2027

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**Analyzing the retraining frequency of global forecasting models in the presence of structural breaks**  
Zanotti, M., Crone, S. F., arXiv
- 2027

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**Sustainable forecasting with artificial intelligence: The role of retraining frequency**  
Zanotti, M., In M. Hamoudia, E. Spiliotis, & P. Montero-Manso (Eds.), New frontiers in forecasting with artificial intelligence: Methods and applications. Palgrave Macmillan
- 2026

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**Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems**  
Zanotti, M., arXiv
- 2026

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**An evaluation of ensemble strategies for time series anomaly detection**  
Zanotti, M., Zangirolami, V., Pavesi, F., Springer
- 2026

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**A characterization of Gaussian processes over flat tori**  
Pavesi, F., Zanotti, M., Zangirolami, V., Springer
- 2026

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**Safe Exogenous State Reinforcement Learning for water tank system**  
Zangirolami, V., Pavesi, F., Zanotti, M., Springer
- 2026

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**Do Short Exposure and Systematic Risk Exposure Drive Asymmetries in the Disposition Effect?**  
Mazzucchelli, L., Zanotti, M., et al., arXiv
- 2025

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**On the retraining frequency of global models in retail demand forecasting**  
Zanotti, M., Machine Learning with Applications, 22, 100769
- 2025

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**The cost of ensembling: Is it always worth combining?**  
Zanotti, M., arXiv
- 2021

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**dispositionEffect**  
Zanotti, M., Mazzucchelli, L., CRAN, R Software



## CONFERENCES & SEMINARS

- 2026

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**Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems**  
International Symposium on Forecasting  Montreal, Canada

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- A vertical timeline on the left side of the page, marked with years 2026, 2025, and 2021. A light blue vertical line runs through the center of these years, with dark blue dots at each year mark. To the right of the line, text describes presentations and conferences, with location pins indicating where they took place.
- 2026**
    - **An evaluation of ensemble strategies for time series anomaly detection**  
SIS-FENStatS  Rome, Italy
  - 2025**
    - **Do global forecasting models require frequent retraining?**  
International Symposium on Forecasting  Beijin, China
  - 2021**
    - **Computing Disposition Effect on Financial Market Data**  
useR! 2021 Conference  Remote